

1 Report Task 3: Classifier Design and Experimental Choices

This section describes the design choices behind the three classifiers used for CIFAR-10. All models were trained on PCA-reduced features (200 components), which preserve around 94% of the original variance while significantly reducing dimensionality and computational cost. Hyperparameters were selected through cross-validation on the training split only.

1.1 Decision Tree

A grid search was run over split criteria, depths, feature selection strategies, and node-regularisation parameters (72 models, 5-fold CV). Training the full grid took 900 seconds. The best model was shallow and regularised:

Hyperparameter	Value
criterion	gini
max_depth	10
min_samples_leaf	4
min_samples_split	2
max_features	None
splitter	best
ccp_alpha	0.0001

Table 1: Best Decision Tree hyperparameters.

	Train	Test
Decision Tree	0.381	0.315

Table 2: Decision Tree accuracy scores.

During tuning, deeper trees (40, 50, or unlimited depth) were tested but quickly abandoned because they exhibited severe overfitting: training accuracy approached 80%, while test accuracy stayed around 25-30%. To control this, a second small grid was run over `ccp_alpha` values to prune the tree. This reduced the gap between training and test accuracy and produced a more stable model. The chosen value, $\alpha = 0.0001$, achieved the strongest overall validation performance.

Misclassifications The decision tree makes inconsistent predictions, often collapsing many inputs into a few dominant classes such as “truck” or “frog”, even when the true label is visually unrelated (e.g., cat $\rightarrow$ truck, automobile $\rightarrow$ frog). This behaviour suggests that the tree relies heavily on a few PCA directions that create coarse partitions in feature space.

Because the tree uses only axis-aligned splits on individual PCA components, it cannot model the nonlinear feature interactions needed for image recognition. The result is unstable and poorly calibrated predictions.

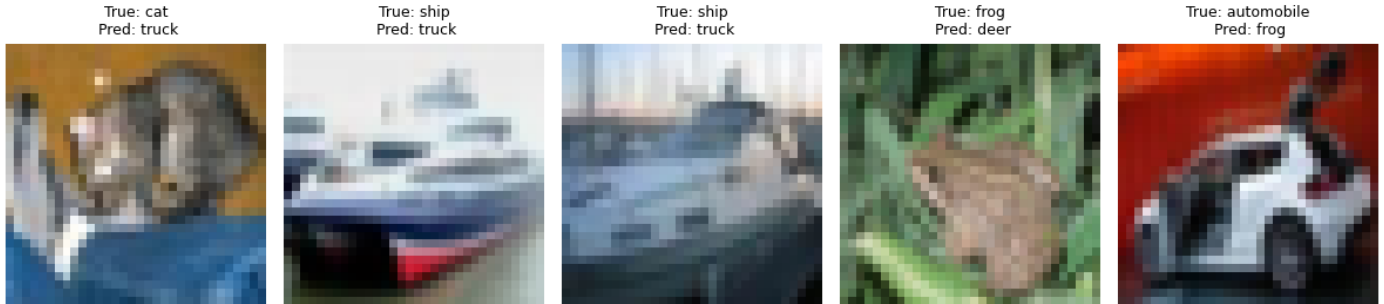


Figure 1: Five misclassified CIFAR-10 test images from the decision tree classifier.

1.2 Ensemble Method (Random Forest)

A Random Forest was used to reduce the high variance of a single tree and to introduce robustness through bootstrap aggregation. Due to computational constraints, a focused grid (36 configurations with 4-fold cross-validation) was used. The optimal hyperparameters were

Hyperparameter	Value
n_estimators	300
max_depth	None
min_samples_leaf	1
min_samples_split	5
max_features	sqrt

Table 3: Selected Random Forest hyperparameters.

	Train	Test
Random Forest	1.000	0.479

Table 4: Random Forest accuracy scores.

Although Random Forest improved substantially over a single decision tree, the model overfits: training accuracy was 1.0 while test accuracy remained below 0.50. This is expected because each tree is grown to full depth, allowing the forest

to memorise PCA components rather than learn generalisable structures.

Misclassifications Misclassifications clustered around similar-looking categories, indicating that although the forest captures more structure than a single tree, it still cannot extract high-level visual features.

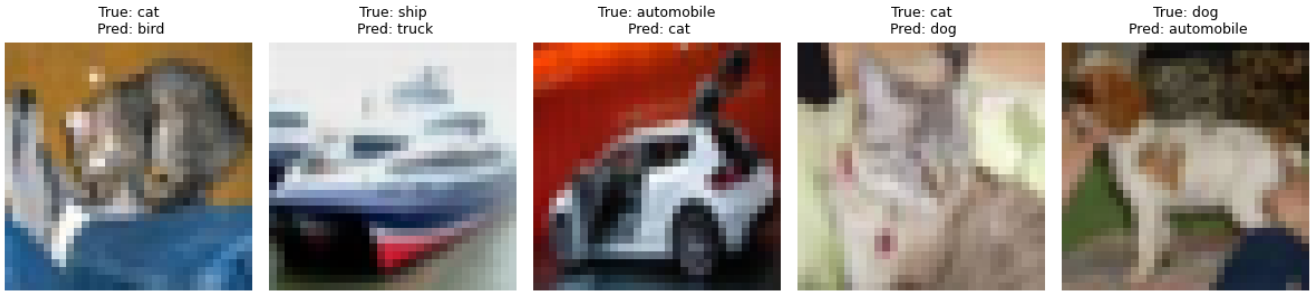


Figure 2: Five misclassified test images from the Random Forest classifier.

1.3 Support Vector Machine

Both linear and RBF kernels were explored. Linear SVMs were computationally heavy and plateaued at roughly 0.41 validation accuracy. Because the `gamma` parameter is only meaningful for nonlinear kernels, the hyperparameter grid was split into two dictionaries: one for linear SVMs (tuning only C), and one for RBF SVMs (tuning both C and `gamma`). This prevented irrelevant parameters being passed to the linear model.

RBF kernels performed substantially better overall. A focused grid over $C \in \{0.1, 1, 10\}$ and `gamma` $\in \{\text{scale}, \text{auto}\}$ identified the optimal model.

Hyperparameter	Value
kernel	rbf
C	10
gamma	scale

Table 5: Selected SVM hyperparameters.

	Train	Test
SVM (RBF)	0.953	0.568

Table 6: SVM accuracy scores.

This SVM achieved the strongest overall performance. However, it clearly overfits: its training accuracy is far higher than test accuracy.

Misclassifications Misclassifications typically occurred on low-resolution images with overlapping shapes or ambiguous colours (e.g., frog $\rightarrow$ deer, truck $\rightarrow$ automobile). These reflect the limitations of shallow, PCA-based features when applied to complex natural images.



Figure 3: Five misclassified CIFAR-10 test images using the tuned SVM classifier.

2 Report Task 4: Comparison of Classifier Performance

This section compares the three classifiers evaluated on PCA-reduced CIFAR-10 data. The Decision Tree achieved the lowest test accuracy, followed by the Random Forest, while the SVM with an RBF kernel performed best. These results align with expectations given the structure and difficulty of CIFAR-10, which contains high intra-class variation and visually overlapping categories.

Model	Train Acc	Test Acc	Train Time (s)
Decision Tree	0.381	0.315	10.14
Random Forest	1.000	0.479	223.54
SVM (RBF)	0.953	0.568	452.06

Table 7: Training/test accuracy and runtime comparison for the three classifiers.

2.1 Model Complexity and Bias-Variance Trade-offs

The Decision Tree is computationally cheap (training in about 10 seconds), but its expressive power is too limited for image data. The model shows underfitting and moderate variance: training accuracy is only 38% and test accuracy is around 31%. The axis-aligned splits cannot capture curved decision regions present in PCA features.

The Random Forest improves accuracy by reducing variance through ensembling, but the cost is high: even a single fit takes over 200 seconds. Despite the large model size, it overfits heavily, achieving perfect memorisation of the training set while test accuracy remains below 50%. This indicates the forest primarily learns low-level colour and texture cues rather than robust object-level structure, and PCA further removes spatial information, limiting tree-based models.

The RBF SVM offers the best trade-off. Its runtime is the highest (around 450 seconds for the final model), but it overfits far less than the Random Forest. The kernel smooths non-linear class boundaries, allowing the SVM to capture meaningful structure without memorising the training data as aggressively. Despite the computational cost, it achieves the strongest performance among the three.

2.2 Suitability for High-Dimensional Image Data

CIFAR-10 images, even after PCA, remain highly complex. Linear decision boundaries (Decision Tree, linear SVM) are insufficient. Tree-based models struggle because pixel-level patterns are not well captured by the random subspace method. In contrast, the SVM’s kernel machinery better captures class structure in the PCA-transformed feature space.

2.3 Insights From Misclassifications

The Decision Tree produces the most unstable errors, often predicting entirely unrelated classes, showing that axis-aligned splits fail to capture meaningful structure.

The Random Forest reduces this instability but still relies heavily on shallow cues such as colour or background texture, leading to confusions when images share similar low-level patterns.

The RBF SVM makes the most coherent errors: its misclassifications usually occur on genuinely ambiguous or low-quality images where class boundaries are visually unclear. Compared with the tree models, it avoids extreme mistakes.

Overall, these patterns reinforce the performance hierarchy: the Decision Tree struggles to learn useful structure, the Random Forest improves but still overfits shallow cues, and the SVM provides the most stable and meaningful predictions.